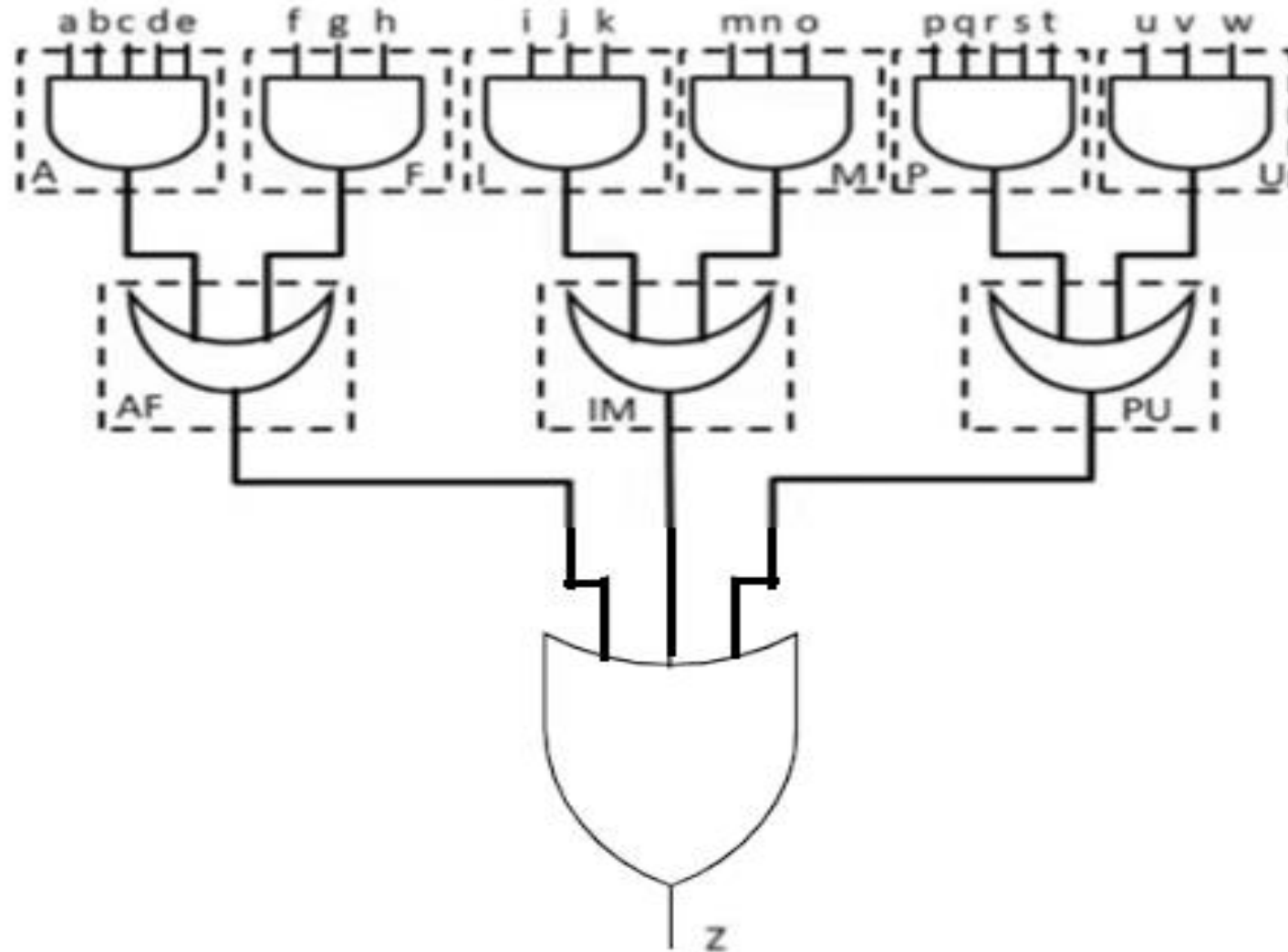


Revision-3

Using the Chortle-crf technique, map the circuit given below to 5-input LUTs. Find the minimum number of 5-input LUTs required and draw the LUT mapped diagram.



Show the generic block level implementation (label all inputs, outputs) of any arbitrary 6-input function $F(x_0, x_1, x_2, x_3, x_4, x_5)$ (i.e. any 6-input function) using minimum number of elements under the following constraints:

- (i) Only 3-input LUTs are available
- (ii) 3-input LUTs and Seven 2:1 multiplexers are available
- (iii) Only 4-input LUTs are available

Determine the number of LUTs required for each of the above three cases.

Case (i)

- We want to implement $f(x_0, x_1, x_2, x_3, x_4, x_5)$ using **only 3-input LUTs**.
- Each LUT can implement any function of **3 inputs**.
- Function has **6 inputs** → Use **Shannon decomposition** (also called recursive multiplexer decomposition).

- **Shannon decomposition**

- For a function $f(x_0, x_1, x_2, x_3, x_4, x_5)$ pick one variable, say x_5 . Then:

$$f(x_0, \dots, x_5) = (\overline{x_5} \cdot f(x_0, x_1, x_2, x_3, x_4, 0)) + (x_5 \cdot f(x_0, x_1, x_2, x_3, x_4, 1))$$

When $x_5 = 0$, the function reduces to $f(\dots, 0)$

When $x_5 = 1$, the function reduces to $f(\dots, 1)$

- Then use a 2:1 MUX with **select = x5** to choose between the two.
- This reduces the number of variables by 1 each time. If we repeat, we eventually reach 3-variable functions that can be directly implemented by a 3-input LUT.